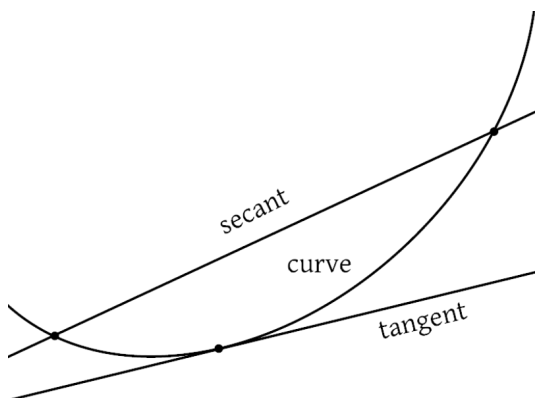


Chapter 17 - Applications of Derivatives

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#Geometric Interpretation of Derivative



Secant	Tangent
The straight line which exactly touches a curve at two points is called secant to the curve.	The straight line which exactly touches a curve at one point is called tangent to the curve.
Slope of secant: $m_{sec} = \frac{\Delta y}{\Delta x}$	Slope of tangent: $m_{tan} = \frac{dy}{dx}$

Relation between the slope of secant and tangent

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

which is the derivative of the function.

Hence, we can say the derivative of a function at a specific point is the slope of the tangent at that point.

#Applications of 1st Derivative

1. To find the stationary/critical point

Stationary/Critical point: The point at which the function is neither increasing nor decreasing. i.e. the tangent is parallel to the X-axis.

Procedure to find the stationary point:

1. Find $f'(x)$
2. Solve for $f'(x) = 0$
3. Suppose $x = a$ is got then it is the stationary point.

2. To determine the monotonicity of a function

Monotonicity: The property of a function whether it is increasing or decreasing.

1. **Increasing function:** If for every $x_1, x_2 \in (a, b)$, $x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$ then the function $y = f(x)$ is said to be increasing in the interval (a, b) . Alternatively, If $x \uparrow \Rightarrow y \uparrow$ then function $y = f(x)$ is said to be increasing function.
2. **Decreasing function:** If for every $x_1, x_2 \in (a, b)$, $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$ then the function $y = f(x)$ is said to be decreasing in the interval (a, b) . Alternatively, If $x \uparrow \Rightarrow y \downarrow$ then function $y = f(x)$ is said to be decreasing function.

Procedure to determine the monotonicity of a function:

1. Find $f'(x)$
2. Check
 - a. If $f'(x) > 0$ then $f(x)$ is increasing.
 - b. If $f'(x) < 0$ then $f(x)$ is decreasing.

3. To find the global (absolute) extreme values

Global/Absolute extreme value: The smallest or greatest value of a function over its entire domain.

1. **Global/Absolute minima:** The smallest value of the function among all x belonging to its domain.
2. **Global/Absolute maxima:** The greatest value of the function among all x belonging to its domain

Procedure to find the global maxima and minima:

1. Find $f'(x)$
2. Solve for $f'(x) = 0$
3. Suppose $x = a, b, c, \dots$ is got then put them in $f(x)$
4. For extreme value
 - a. The least value of $f(x)$ = absolute minimum value.
 - b. The greatest value of $f(x)$ = absolute maximum value.

#Applications of 2nd Derivative

1. To find the point of inflection

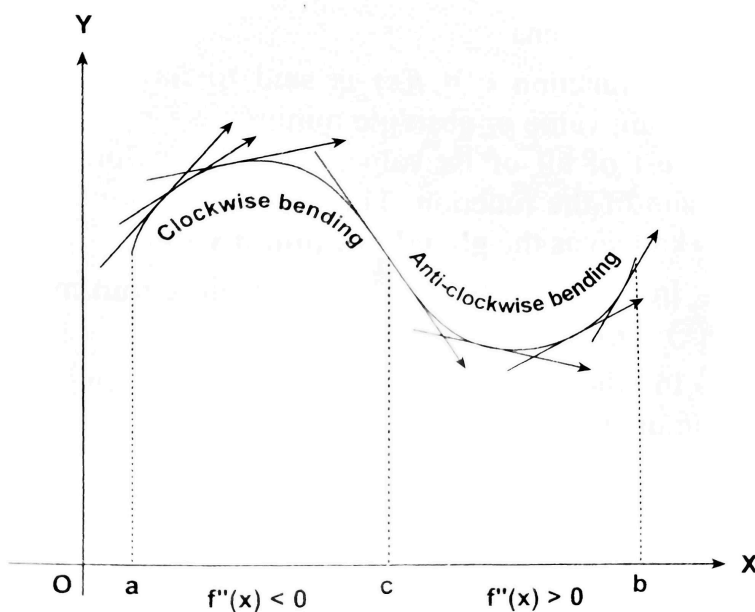
Point of inflection: The point which separates the concave upward and downward portions of the graph.

Procedure to find the point of inflection:

1. Find $f'(x)$
2. Also find $f''(x)$
3. Solve for $f''(x) = 0$
4. Suppose $x = a$ is got then (if $f'''(a) \neq 0$ then) it is the point of inflection

2. To determine the concavity of a function in an interval

Concavity: The property of a function whether it is concave upward or downward.



1. **Concave upward:** The portion in which the function is in clockwise bending.
2. **Concave downward:** The portion in which the function is in anti-clockwise bending.

Procedure to determine the concavity of a function in an interval:

1. Find $f'(x)$
2. Also find $f''(x)$
3. Solve for $f''(x) = 0$
4. Suppose $x = a, b, c, \dots$ is got then make interval from them and one by one plug the some values in between the intervals in $f''(x)$
 - a. If $f''(x) > 0$ then it is concave upward in the related interval.
 - b. If $f''(x) < 0$ then it is concave downward in the related interval.

3. To find the local (relative) extreme values in an interval

Local/Relative extreme value: The smallest or greatest value of function within a specified interval.

1. **Local/Relative minima:** The smallest value of the function among all points of inflection and the end points in an interval.
2. **Local/Relative maxima:** The greatest value of the function among all points of inflection and the end points in an interval.

Procedure to find the maxima and minima:

1. Find $f'(x)$
2. Also find $f''(x)$
3. Solve for $f'(x) = 0$
4. Suppose $x = a, b, c, \dots$ is got then put them in $f''(x)$
 - a. If $f''(x) < 0$ then it is the maximum value.
 - b. If $f''(x) > 0$ then it is the minimum value.